Кочеткова М. П. Интегралы/часть2

Часть 2

8.2.33

$$\int \cos(6x+1)\,dx = \begin{bmatrix} y = 6x+1 \Rightarrow dy = d(6x+1) = \\ (6x+1)'_x\,dx = 6dx \Rightarrow dx = \frac{1}{6}dy \end{bmatrix} =$$

$$= \int \cos y \cdot \frac{1}{6}dy = \frac{1}{6}\int \cos y\,dy = \frac{1}{6}\sin y + C = \frac{\sin(6x+1)}{6} + C$$

8.2.34

$$\int \frac{dx}{\sqrt[3]{(5x-2)^4}} = \int (5x-2)^{-\frac{4}{3}}dx = \begin{bmatrix} y = 5x-2 = dy = d(5x-2) = \\ = (5x-2)'_x\,dx = 5dx \Rightarrow dx = \frac{1}{5}dy \end{bmatrix} =$$

$$= \frac{1}{5}\cdot\frac{y^{-\frac{4}{3}+1}}{-\frac{4}{3}+1} + C = \frac{1}{5}\cdot -\frac{3}{\sqrt[3]{y}} + C = -\frac{3}{5\cdot\sqrt[3]{y}} + C = -\frac{3}{5\sqrt[3]{5x-2}} + C$$

8.2.35

$$\int \frac{\sqrt{tg\,x}\,dx}{\cos^2 x} = \begin{bmatrix} y = tg\,x \Rightarrow dy = d(tg\,x) = \\ = (tg\,x)'_x\,dx = \frac{1}{\cos^2 x}dx \end{bmatrix} =$$

$$= \int \sqrt{y}\,dy = \frac{y^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C = \frac{y^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{2\sqrt{y^3}}{3} + C =$$

$$= \frac{2\sqrt{tg^3 x}}{3} + C$$

8.2.36

$$\int \frac{e^x\,dx}{e^{2x}+9} = \begin{bmatrix} y = e^x \Rightarrow dy = d(e^x) = \\ = (e^x)'_x\,dx = e^x dx \end{bmatrix} = \int \frac{dy}{y^2+3^2} =$$

$$= \frac{1}{3}\,arctg\,\frac{y}{3} + C = \frac{1}{3}\cdot arctg\,\frac{e^x}{3} + C$$

8.2.37

$$\int \frac{x^5\,dx}{\sqrt{x^6+7}} = \begin{bmatrix} y = x^6+7 \Rightarrow dy = d(x^6+7) = \\ = (x^6+7)'_x\,dx = 6x^5dx \end{bmatrix} = \int \frac{dy}{6\sqrt{y}} = \frac{1}{6}\int\frac{dy}{\sqrt{y}} =$$

$$= \frac{1}{6}\cdot 2\sqrt{y} + C = \frac{\sqrt{x^6+7}}{3} + C$$

8.2.38

$$\int \frac{dx}{\arccos x \cdot \sqrt{1-x^2}} = \left[\begin{array}{l} y = \arccos x \Rightarrow dy = d(\arccos x) = \\ = (\arccos x)'_x\, dx = -\frac{1}{\sqrt{1-x^2}}\, dx \end{array}\right] = \int -\frac{dy}{y} =$$

$$= -\ln|y| + C = -\ln|\arccos x| + C$$

8.2.39

$$\int \frac{(2x+3)\,dx}{(x^2+3x-1)^4} = \left[\begin{array}{l} y = x^2+3x-1 \Rightarrow dy = d(x^2+3x-1) = \\ = (x^2+3x-1)'_x\, dx = (2x+3)\,dx \end{array}\right] = \int \frac{dy}{y^4} = \int y^{-4}\,dy =$$

$$= \frac{y^{-4+1}}{-4+1} + C = -\frac{1}{3y^3} + C = -\frac{1}{3(x^2+3x-1)^3} + C$$

8.2.40

$$\int \cos^{11}2x \sin 2x\, dx = \left[y = \cos 2x \Rightarrow dy = d(\cos 2x) = (\cos 2x)'_x\, dx = -2\sin 2x\, dx\right] =$$

$$= \int -\frac{y^{11}}{2}\,dy = -\frac{1}{2}\int y^{11}\,dy = -\frac{1}{2}\cdot\frac{y^{11+1}}{11+1} + C = -\frac{1}{2}\cdot\frac{y^{12}}{12} + C = -\frac{y^{12}}{24} + C =$$

$$= -\frac{\cos^{12}2x}{24} + C$$

8.2.41

$$\int \frac{7^{\sqrt{x}}\,dx}{\sqrt{x}} = \left[\begin{array}{l} y = \sqrt{x} \Rightarrow dy = d(\sqrt{x}) = \\ = (\sqrt{x})'_x\, dx = \frac{dx}{2\sqrt{x}} \end{array}\right] = \int \frac{7^y \cdot 2 \cdot \sqrt{x}}{\sqrt{x}}\,dy = \int 2\cdot 7^y\,dy =$$

$$= 2\int 7^y\,dy = 2\cdot\frac{7^y}{\ln 7} + C = \frac{2\cdot 7^{\sqrt{x}}}{\ln 7} + C$$

8.2.42

$$\int \frac{e^{\frac{1}{x}}}{x^2}\,dx = \left[\begin{array}{l} y = \frac{1}{x} \Rightarrow dy = d\left(\frac{1}{x}\right) = \\ = \left(\frac{1}{x}\right)'_x dx = -\frac{1}{x^2}\,dx \end{array}\right] = \int -e^y\,dy = -\int e^y\,dy =$$

$$= -e^y + C = -e^{\frac{1}{x}} + C$$

8.2.43

$$\int \frac{\ln 5x \, dx}{x} = \left[\begin{matrix} y = \ln 5x \Rightarrow dy = d(\ln 5x) = \\ = (\ln 5x)'_x \, dx = \frac{5}{5x} \, dx = \frac{dx}{x} \end{matrix}\right] = \int y \, dy =$$

$$= \frac{y^2}{2} + C = \frac{(\ln 5x)^2}{2} + C$$

8.2.44

$$\int ctg\, x \, dx = \int \frac{\cos x}{\sin x} dx = \left[\begin{matrix} y = \sin x \Rightarrow dy = d(\sin x) = \\ = (\sin x)'_x \, dx = \cos x \, dx \end{matrix}\right] =$$

$$\int \frac{dy}{y} = \ln|y| + C = \ln|\sin x| + C$$

8.2.45

$$\int 4x \cdot \sqrt[3]{x^2+8} \, dx = 4 \int x \sqrt[3]{x^2+8} \, dx = \left[y = x^2+8 \Rightarrow dy = d(x^2+8) = (x^2+8)'_x \, dx = \right.$$

$$\left. = 2x \, dx \right] = 4 \int \frac{\sqrt[3]{y}}{2} dy = 2 \int y^{\frac{1}{3}} dy = 2 \cdot \frac{y^{4/3}}{\frac{4}{3}} + C = \frac{3}{2} (x^2+8)^{\frac{4}{3}} + C =$$

$$= \frac{3}{2} \cdot (x^2+8) \cdot \sqrt[3]{(x^2+8)} + C$$

8.2.46

$$\int \frac{\cos x \, dx}{\sin^2 x} = \left[\begin{matrix} y = \sin x \Rightarrow dy = d(\sin x) = \\ = (\sin x)'_x \, dx = \cos x \, dx \end{matrix}\right] = \int \frac{dy}{y^2} = -\frac{1}{y} + C =$$

$$= -\frac{1}{\sin x} + C$$

8.2.47

$$\int tg\, 2x \, dx = \left[y = 2x \Rightarrow dy = d(2x) = (2x)'_x \, dx = 2dx \right] = \int \frac{tg\, y}{2} dy =$$

$$= \frac{1}{2} \int \frac{\sin y}{\cos y} dy = \left[\begin{matrix} t = \cos y \Rightarrow dt = d(\cos y) = \\ = (\cos y)'_y \, dy = -\sin y \, dy \end{matrix}\right] = \frac{1}{2} \int -\frac{dt}{t} = -\frac{1}{2} \int \frac{dt}{t} =$$

$$= -\frac{1}{2} \ln|t| + C = -\frac{\ln|\cos y|}{2} + C = -\frac{\ln|\cos 2x|}{2} + C$$

8.2.48

$$\int \frac{x\,dx}{x^4+1} = \left[\begin{array}{l} y = x^2 \Rightarrow dy = d(x^2) = \\ = (x^2)'_x\,dx = 2x\,dx \end{array} \right] = \int \frac{1}{2} \cdot \frac{dy}{y^2+1} = \frac{1}{2}\int \frac{dy}{y^2+1} =$$

$$= \frac{1}{2}\,\text{arctg}\,y + C = \frac{\text{arctg}(x^2)}{2} + C$$

8.2.49

$$\int e^{-x^3} \cdot x^2\,dx = \left[y = e^{-x^3} \Rightarrow dy = d(e^{-x^3}) = (e^{-x^3})'_x\,dx = e^{-x^3}\cdot(-3x^2)\,dx \right] =$$

$$= \int -\frac{1}{3}\,dy = -\frac{y}{3} + C = -\frac{e^{-x^3}}{3} + C = -\frac{1}{3e^{x^3}} + C$$

8.2.50

$$\int \frac{x^2\,dx}{\sqrt{x^6-4}} = \left[\begin{array}{l} y = x^3 \Rightarrow dy = d(x^3) = \\ = (x^3)'_x\,dx = 3x^2\,dx \end{array} \right] = \frac{1}{3}\cdot\int \frac{dy}{\sqrt{y^2-4}} =$$

$$= \frac{1}{3}\cdot \ln\left|y + \sqrt{y^2-4}\right| + C = \frac{1}{3}\cdot \ln\left|x^3 + \sqrt{x^6-4}\right| + C$$

8.2.51

$$\int \left(8\cos\frac{x}{3} - 5\right)^2 \sin\frac{x}{3}\,dx = \left[\begin{array}{l} y = 8\cos\frac{x}{3} - 5 \Rightarrow dy = d\left(8\cos\frac{x}{3} - 5\right) = \\ = \left(8\cos\frac{x}{3} - 5\right)'_x dx = 8\cdot\left(-\sin\frac{x}{3}\right)\cdot\frac{1}{3}\,dx \end{array} \right] =$$

$$= -\frac{3}{8}\int y^2\,dy = -\frac{3}{8}\cdot\frac{y^3}{3} + C = -\frac{3\cdot\left(8\cos\frac{x}{3} - 5\right)^3}{8\cdot 3} + C =$$

$$= -\frac{\left(8\cos\frac{x}{3} - 5\right)^3}{8} + C$$

8.2.52

$$\int \frac{(3x^2 - 2x + 7)\,dx}{\sqrt{x^3 - x^2 + 7x - 2}} = \left[\begin{array}{l} y = (x^3 - x^2 + 7x - 2) \Rightarrow dy = d(x^3 - x^2 + 7x - 2) = \\ = (x^3 - x^2 + 7x - 2)'_x\,dx = (3x^2 - 2x + 7)\,dx \end{array} \right] =$$

$$= \int \frac{dy}{\sqrt{y}} = 2\sqrt{y} + C = 2\cdot\sqrt{x^3 - x^2 + 7x - 2} + C$$

8.2.53

$$\int x(2x+1)^{35}dx = \left[y = 2x+1 \Rightarrow dy = d(2x+1) = (2x+1)'_x dx = 2dx\right] =$$

$$= \int \frac{\frac{y-1}{2}\cdot y^{35}}{2}dy = \int \frac{(y-1)\cdot y^{35}}{4}dy = \int \frac{y^{36}-y^{35}}{4}dy = \frac{1}{4}\int (y^{36}-y^{35})dy =$$

$$= \frac{1}{4}\left(\int y^{36}dy - \int y^{35}dy\right) + C = \frac{1}{4}\cdot\left(\frac{y^{37}}{37} - \frac{y^{36}}{36}\right) + C = \frac{y^{37}}{148} - \frac{y^{36}}{144} + C =$$

$$= \frac{(2x+1)^{37}}{148} - \frac{(2x+1)^{36}}{144} + C$$

8.2.54

$$\int (x-2)\sqrt{x+4}\,dx = \int (x+4-4-2)\sqrt{x+4}\,dx = \int (x+4-6)\sqrt{x+4}\,dx =$$

$$= \left[y = x+4 \Rightarrow dy = d(x+4) = (x+4)'_x dx = dx\right] = \int (y-6)\sqrt{y}\,dy =$$

$$= \int (y\cdot\sqrt{y} - 6\sqrt{y})dy = \int y^{\frac{3}{2}}dy - 6\int y^{\frac{1}{2}}dy =$$

$$= \frac{y^{\frac{3}{2}+1}}{\frac{3}{2}+1} - 6\cdot\frac{y^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C = \frac{2y^{\frac{5}{2}}}{5} - \frac{12y^{\frac{3}{2}}}{3} + C =$$

$$= \frac{2y^2\sqrt{y}}{5} - 4y\sqrt{y} + C = \frac{2\cdot(x+4)^2\cdot\sqrt{x+4}}{5} - 4(x+4)\cdot\sqrt{x+4} + C$$

8.2.55.

$$\int \frac{3\sqrt{x} - 2\cos\frac{1}{x^2}}{x^3}dx = \int \frac{3\sqrt{x}}{x^3}dx - \int \frac{2\cos\frac{1}{x^2}}{x^3}dx =$$

$$= 3\int x^{-\frac{5}{2}}dx - \int \frac{2\cos\frac{1}{x^2}}{x^3}dx = \left[\begin{matrix} y = \cos\frac{1}{x^2} \Rightarrow dy = d(\cos\frac{1}{x^2}) = \\ = (\cos\frac{1}{x^2})'_x dx = -\sin\frac{1}{x^2}\cdot(-2)\cdot\frac{1}{x^3}\cdot dx \end{matrix}\right] = \left[\begin{matrix}\sin\frac{1}{x^2} = \\ \sqrt{1-\cos\frac{1}{x^2}}\end{matrix}\right] =$$

$$= 3\int x^{-\frac{5}{2}}dx - \int \frac{y}{\sqrt{1-y}}dy = \left[\begin{matrix} t = 1-y \Rightarrow dt = d(1-y) = \\ = (1-y)'_x dy = -dy \end{matrix}\right] = 3\int x^{-\frac{5}{2}}dx - \int -\frac{1-t}{\sqrt{t}}dt =$$

$$= \left[\begin{matrix} 2)\ \int -\frac{1-t}{\sqrt{t}}dt = \int \frac{t-1}{\sqrt{t}}dt = \int t^{\frac{1}{2}}dt - \int \frac{1}{\sqrt{t}}dt = \frac{2t\sqrt{t}}{3} - 2\sqrt{t} + C \\ 1)\ \int x^{-\frac{5}{2}}dx = -\frac{2}{3x\sqrt{x}} + C \end{matrix}\right] =$$

$$= -\frac{6}{3x\sqrt{x}} - \frac{2t\sqrt{t}}{3} + 2\sqrt{t} + C = -\frac{6}{3x\sqrt{x}} - \frac{2(1-y)\sqrt{1-y}}{3} +$$

$$+ 2\sqrt{1-y} + C = -\frac{6}{3x\sqrt{x}} - \frac{2(1-\cos\frac{1}{x^2})\sqrt{1-\cos\frac{1}{x^2}}}{3} + 2\sqrt{1-\cos\frac{1}{x^2}} + C$$

8.2.56.

$$\int \frac{7x+2}{\sqrt{x^2+10}}dx = 7\int\frac{x}{\sqrt{x^2+10}}dx + 2\int\frac{dx}{\sqrt{x^2+10}} =$$

$$= \left[\begin{matrix} y = x^2+10 \Rightarrow dy = d(x^2+10) = \\ = (x^2+10)'_x dx = 2x\,dx \end{matrix}\right] = \frac{7}{2}\int\frac{dy}{\sqrt{y}} + 2\int\frac{dx}{\sqrt{x^2+10}} =$$

$$= 7\sqrt{y} + 2\ln|x+\sqrt{x^2+10}| + C =$$

$$= 7\sqrt{x^2+10} + 2\ln|x+\sqrt{x^2+10}| + C.$$

8.2.57.

$$\int\frac{dx}{e^x+e^{-x}} = \int\frac{dx}{e^x+\frac{1}{e^x}} = \int\frac{dx}{\frac{e^{2x}+1}{e^x}} = \int\frac{e^x}{e^{2x}+1}dx =$$

$$= \left[\begin{matrix} y = e^x \Rightarrow dy = d(e^x) = \\ = (e^x)'_x dx = e^x dx \end{matrix}\right] = \int\frac{dy}{y^2+1} = \operatorname{arctg} y + C =$$

$$= \operatorname{arctg} e^x + C$$

8.2.58.

$$\int\frac{x+8}{x^2+3}dx = \int\frac{x}{x^2+3}dx + 8\int\frac{dx}{x^2+3} = \left[\begin{matrix} y = x^2+3 \Rightarrow dy = d(x^2+3) \\ = (x^2+3)'_x dx = 2x\,dx \end{matrix}\right] =$$

$$= \frac{1}{2}\int\frac{dy}{y} + 8\int\frac{dx}{x^2+(\sqrt{3})^2} = \frac{\ln|y|}{2} + \frac{8}{\sqrt{3}}\operatorname{arctg}\frac{x}{\sqrt{3}} + C =$$

$$= \frac{\ln|x^2+3|}{2} + \frac{8}{\sqrt{3}}\operatorname{arctg}\frac{x}{\sqrt{3}} + C = \frac{\ln(x^2+3)}{2} + \frac{8}{\sqrt{3}}\operatorname{arctg}\frac{x}{\sqrt{3}} + C$$

8.2.59

$$\int \frac{x + 4\sqrt{\arcsin x}}{\sqrt{1-x^2}}\,dx = \int \frac{x\,dx}{\sqrt{1-x^2}} + 4\int \frac{(\arcsin x)^{\frac{1}{2}}}{\sqrt{1-x^2}} =$$

$$= \Big[\ 1)\ \int \frac{x}{\sqrt{1-x^2}}\,dx = \left[\begin{matrix} y = \sqrt{1-x^2} \Rightarrow dy = d(\sqrt{1-x^2}) = \\ = (\sqrt{1-x^2})'_x\,dx = -\frac{x}{\sqrt{1-x^2}}\,dy \end{matrix}\right] = \int -dy = -y + C$$

$$2)\ \int \frac{(\arcsin x)^{\frac{1}{2}}}{\sqrt{1-x^2}}\,dx = \left[\begin{matrix} t = \arcsin x \Rightarrow dt = d(\arcsin x) = \\ = (\arcsin x)'_x\,dx = \frac{1}{\sqrt{1-x^2}}\,dx \end{matrix}\right] = \int t^{\frac{1}{2}}\,dt =$$

$$= \frac{2t\sqrt{t}}{3} + C\ \Big] =$$

$$= -y + 4\cdot\frac{2t\sqrt{t}}{3} + C = -\sqrt{1-x^2} + \frac{8\arcsin x\sqrt{\arcsin x}}{3} + C$$

8.2.60

$$\int \frac{1-6x}{(x+1)(x-1)}\,dx = \int \frac{dx}{x^2-1^2} - 6\int \frac{x}{x^2-1^2}\,dx = \left[\begin{matrix} y = x^2-1 \Rightarrow dy = d(x^2-1) = \\ = (x^2-1)'_x dx = 2x\,dx \end{matrix}\right] =$$

$$= \int \frac{dx}{x^2-1^2} - 3\int \frac{dy}{y} = \left[\begin{matrix} 1)\ \int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C \\ 2)\ \int \frac{dx}{x} = \ln|x| + C \end{matrix}\right] =$$

$$= \frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| - 3\ln|y| + C = \frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| - 3\ln|x^2-1| + C$$

8.2.61

$$\int (\cos^2 x - \sin^2 x)\sqrt[3]{1+\sin(2x)}\,dx = \int \cos 2x\,\sqrt[3]{1+\sin(2x)}\,dx =$$

$$= \left[y = 1+\sin(2x) \Rightarrow dy = d(1+\sin(2x)) = (1+\sin(2x))'_x\,dx = 2\cos(2x)dx\right] = \frac{1}{2}\int y^{\frac{1}{3}}dy + C =$$

$$= \frac{1}{2}\cdot\frac{3y^{\frac{4}{3}}}{4} + C = \frac{3y\sqrt[3]{y}}{8} + C = \frac{3(1+\sin(2x))\sqrt[3]{1+\sin(2x)}}{8} + C$$

8.2.62

$$\int \frac{e^{\operatorname{tg}x} - 7\sin x + 5\sin 2x}{\cos^2 x}\,dx = \int \left(\frac{e^{\operatorname{tg}x}}{\cos^2 x} - \frac{7\sin x}{\cos^2 x} + \frac{5\sin 2x}{\cos^2 x}\right)dx =$$

$$= \int \frac{e^{tgx}}{\cos^2 x}dx - 7\int \frac{\sin x}{\cos^2 x}dx + 5\int \frac{2\sin x\cos x}{\cos^2 x}dx =$$

$$= \int \frac{e^{tgx}}{\cos^2 x}dx - 7\int \frac{\sin x}{\cos^2 x}dx + 10\int \frac{\sin x}{\cos x}dx =$$

$$= \Big[\ 1)\ \int \frac{e^{tgx}}{\cos^2 x}dx = \left[\begin{matrix} y = e^{tgx} \Rightarrow dy = d(e^{tgx}) = \\ = (e^{tgx})'_x dx = \frac{e^{tgx}}{\cos^2 x}dx \end{matrix}\right] = \int dy = y + C = e^{tgx} + C$$

$$2)\ \int \frac{\sin x}{\cos^2 x}dx = \left[\begin{matrix} t = \cos x \Rightarrow dt = d(\cos x) = \\ = (\cos x)'_x dx = -\sin x\, dx \end{matrix}\right] = \int -\frac{dt}{t^2} = \frac{1}{t} + C = \frac{1}{\cos x} + C$$

$$3)\ \int \frac{\sin x}{\cos x}dx = \int tg x\, dx = -\ln|\cos x| + C \quad \Big] =$$

$$= e^{tgx} - \frac{7}{\cos x} - 10\ln|\cos x| + C.$$

8.2.63

$$\int \sqrt{16 - x^2}\,dx = \left[x = 4\sin t \Rightarrow dx = d(4\sin t) = (4\sin t)'_t dt = 4\cos t\, dt\right] =$$

$$= \int \sqrt{16 - 16\sin^2 t}\cdot 4\cos t\, dt = 4\int \sqrt{16(1-\sin^2 t)}\cos t\, dt = 4\int \sqrt{16\cos^2 t}\cos t\, dt =$$

$$= 4\int 4\cos t\cos t\, dt = 16\int \cos^2 t\, dt = 16\int \frac{1+\cos 2t}{2}dt = 8\int 1 + \cos 2t\, dt =$$

$$= 8\left(\int dt + \int \cos 2t\, dt\right) = \left[\begin{matrix} 1)\ \int dt = t + C \\ 2)\ \int \cos 2t\, dt = \frac{1}{2}\sin 2t + C \\ (2.1)\ \int \cos x\, dx = \sin x + C \\ (2.2)\ \int f(ax+b)dx = \frac{1}{a}\cdot F(ax+b) + C \end{matrix}\right] = 8\left(t + \frac{\sin 2t}{2}\right) + C =$$

$$= 8\arcsin\frac{x}{4} + 4\sin\left(2\arcsin\frac{x}{4}\right) + C =$$

$$= 8\arcsin\frac{x}{4} + 8\sin\left(\arcsin\frac{x}{4}\right)\cos\left(\arcsin\frac{x}{4}\right) + C =$$

$$= 8\arcsin\frac{x}{4} + 8\cdot\frac{x}{4}\cdot\sqrt{1-\left(\frac{x}{4}\right)^2} + C = 8\arcsin\frac{x}{4} + \frac{2x\sqrt{16-x^2}}{4} + C =$$

$$= 8\arcsin\frac{x}{4} + \frac{x\sqrt{16+x^2}}{2} + C$$

8.2.64

$$\int \frac{dx}{1+\sqrt{x}} = \begin{bmatrix} x = \sin t \Rightarrow dx = d(\sin t) = \\ = (\sin t)'_t\, dt = \cos t\, dt \end{bmatrix} = \int \frac{\cos t\, dt}{1+\sqrt{\sin t}} = \int \frac{\cos t\, dt}{1+\sin t^{\frac{1}{2}}}$$

$$= \begin{bmatrix} v = 1+\sin t^{\frac{1}{2}} \Rightarrow dv = d(1+\sin t^{\frac{1}{2}}) = \\ = (1+\sin t^{\frac{1}{2}})'_t\, dt = \frac{1}{2}\sin t^{-\frac{1}{2}} \cos t\, dt \end{bmatrix} = \int \frac{2\sin t^{\frac{1}{2}}}{1+\sin t^{\frac{1}{2}}}\, dv = \int \frac{2(\sin t^{\frac{1}{2}}+1-1)}{1+\sin t^{\frac{1}{2}}}\, dv =$$

$$= \int \frac{2(v-1)}{v}\, dv = 2\left(\int \frac{v}{v}\, dv - \int \frac{dv}{v}\right) = 2\left(\int dv - \int \frac{dv}{v}\right) =$$

$$= \begin{bmatrix} 1)\ \int dv = v + C \\ 2)\ \int \frac{dv}{v} = \ln|v| + C \end{bmatrix} = 2(v - \ln|v|) + C =$$

$$= 2 + 2\sqrt{\sin t} - 2\ln(1+\sqrt{\sin t}) + C = 2\sqrt{x} - 2\ln(1+\sqrt{x}) + C$$

8.2.65

$$\int x\sqrt{x+3}\, dx = [x = t-3 \Rightarrow dx = d(t-3) = (t-3)'_t\, dt = dt] =$$

$$\int (t-3)\sqrt{t-3+3}\, dt = \int (t-3)\sqrt{t}\, dt = \int t^{\frac{3}{2}}\, dt - 3\int t^{\frac{1}{2}}\, dt =$$

$$= \frac{2t^{\frac{5}{2}}}{5} - 3\,\frac{2t^{\frac{3}{2}}}{3} + C = \frac{2t^2\sqrt{t}}{5} - 2t\sqrt{t} + C =$$

$$= \frac{2(x+3)^2\sqrt{x+3}}{5} - 2(x+3)\sqrt{x+3} + C =$$

$$= \frac{2(x^2+6x+9)\sqrt{x+3}}{5} - 2(x+3)\sqrt{x+3} + C$$

8.2.66

$$\int \frac{dx}{(x+1)\sqrt{x}} = \begin{bmatrix} x = t^2 \Rightarrow dx = d(t^2) = \\ = (t^2)'_t\, dt = 2t\, dt \end{bmatrix} = \int \frac{2t\, dt}{(t^2+1)\sqrt{t^2}} =$$

$$= 2\int \frac{t\, dt}{(t^2+1)\, t} = 2\int \frac{dt}{t^2+1} = 2\,\mathrm{arctg}\, t + C = 2\,\mathrm{arctg}\sqrt{x} + C.$$

8.2.67

$$\int \frac{x\,dx}{\sqrt{1-x}} = \left[\begin{array}{l} x = 1-t \Rightarrow dx = d(1-t) = \\ = (1-t)'_t\, dt = -dt \end{array}\right] = -\int \frac{1-t}{\sqrt{1-1+t}}\,dt =$$

$$= \int \frac{t-1}{\sqrt{t}}\,dt = \int t^{\frac{1}{2}}dt - \int \frac{dt}{\sqrt{t}} = \frac{2t\sqrt{t}}{3} - 2\sqrt{t} + C =$$

$$= \frac{2(1-x)\sqrt{1-x}}{3} - 2\sqrt{1-x} + C$$

8.2.68

$$\int \frac{x^2dx}{\sqrt{1-x^2}} = \left[\begin{array}{l} x = \sin t \Rightarrow dx = d(\sin t) = \\ = (\sin t)'_t\, dt = \cos t\, dt \end{array}\right] = \int \frac{\sin^2 t}{\sqrt{1-\sin^2 t}} \cdot \cos t\, dt =$$

$$= \int \frac{\sin^2 t}{\sqrt{\cos^2 t}} \cdot \cos t\, dt = \int \frac{\sin^2 t}{\cos t} \cdot \cos t\, dt = \int \sin^2 t\, dt = \int \frac{1-\cos 2t}{2}\,dt =$$

$$= \frac{1}{2}\int (1-\cos 2t)\,dt = \frac{1}{2}\left(\int dt - \int \cos 2t\, dt\right) = \left[\begin{array}{l} \text{I. } \int dx = x + C \\ \text{II. } \int \cos x\, dx = \sin x + C \\ \text{III. } \int f(ax+b)\,dx = \frac{1}{a} F(ax+b) + C \end{array}\right] =$$

$$= \frac{1}{2}\left(t - \frac{\sin 2t}{2}\right) + C = \frac{1}{2}\left(\arcsin x - \frac{\sin(2\arcsin x)}{2}\right) + C =$$

$$= \frac{1}{2}\left(\arcsin x - \frac{2\sin(\arcsin x)\cos(\arcsin x)}{2}\right) + C =$$

$$= \frac{1}{2}\left(\arcsin x - x \cdot \sqrt{1-x^2}\right) + C = \frac{\arcsin x - x\sqrt{1-x^2}}{2} + C$$

8.2.69

$$\int x \ln x\, dx = \left[\begin{array}{l} u = \ln x \Rightarrow u' = (\ln x)'_x = \frac{1}{x} \\ v' = x \Rightarrow v = \int v' dx = \int x\, dx = \frac{x^2}{2} \end{array}\right] =$$

$$= \ln x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x}\,dx = \ln x \cdot \frac{x^2}{2} - \int \frac{x}{2}\,dx = \ln x \cdot \frac{x^2}{2} - \frac{1}{2}\int x\, dx =$$

$$= \ln x \cdot \frac{x^2}{2} - \frac{x^2}{4} + C$$

8.2.70

$$\int (2x+3)\cdot \cos x\, dx = \left[\begin{array}{lll} u = 2x+3 & \Rightarrow & u' = (2x+3)'_x = 2 \\ v' = \cos x & \Rightarrow & v = \int v' dx = \int \cos x\, dx = \sin x \end{array}\right] =$$

$$= (2x+3)\cdot \sin x - \int \sin x \cdot 2\, dx = (2x+3)\cdot \sin x - 2\int \sin x\, dx =$$

$$= (2x+3)\cdot \sin x + 2\cos x + C = 2x \sin x + 3\sin x + 2\cos x + C.$$

8.2.71

$$\int x \cdot sh 5x\, dx = \left[\begin{array}{lll} u = x & \Rightarrow & u' = (x)'_x = 1 \\ v' = sh 5x & \Rightarrow & v = \int v' dx = \int sh 5x\, dx = \left[\begin{array}{l} 1) \int sh x\, dx = ch x + C \\ 2) \int f(ax+b) dx = \frac{1}{a} F(ax+b) + C \end{array}\right] \\ & & = \frac{1}{5} ch 5x \end{array}\right] =$$

$$= x \cdot \frac{1}{5} ch 5x - \int \frac{1}{5} ch 5x \cdot 1\, dx = \frac{x}{5} \cdot ch 5x - \frac{1}{5}\int ch 5x\, dx =$$

$$= \left[\begin{array}{l} 1) \int ch x\, dx = sh x + C \\ 2) \int f(ax+b) dx = \frac{1}{a} \cdot F(ax+b) + C \end{array}\right] = \frac{x \cdot ch 5x}{5} - \frac{1}{5} \cdot \frac{1}{5}\, sh 5x + C =$$

$$= \frac{x \cdot ch 5x}{5} - \frac{sh 5x}{25} + C$$

8.2.72

$$\int \frac{x \cdot \cos x\, dx}{\sin^3 x} = \left[\begin{array}{lll} u = x & \Rightarrow & u' = (x)'_x = 1 \\ v' = \frac{\cos x}{\sin^3 x} & \Rightarrow & v = \int \frac{\cos x}{\sin^3 x} dx = -\frac{1}{2\sin^2 x} \end{array}\right] =$$

$$= x \cdot \left(-\frac{1}{2\sin^2 x}\right) - \int -\frac{1}{2\sin^2 x} dx = -\frac{x}{2\sin^2 x} + \frac{1}{2}\int \frac{dx}{\sin^2 x} =$$

$$= -\frac{x}{2\sin^2 x} + \frac{1}{2}(-ctg\, x) + C = -\frac{x}{2\sin^2 x} - \frac{ctg\, x}{2} + C$$

8.2.73

$$\int x^2 \ln x\, dx = \left[\begin{array}{lll} u = \ln x & \Rightarrow & u' = (\ln x)'_x = \frac{1}{x} \\ v' = x^2 & \Rightarrow & v = \int v' dx = \int x^2 dx = \frac{x^3}{3} \end{array}\right] =$$

$$= \ln x \cdot \frac{x^3}{3} - \int \frac{x^3}{3} \cdot \frac{1}{x} dx = \ln x \cdot \frac{x^3}{3} - \frac{1}{3}\int x^2 dx = \ln x \cdot \frac{x^3}{3} - \frac{x^3}{9} + C$$

8.2.74

$$\int (x^2-4x+1)e^{-x}\,dx = \begin{bmatrix} u = x^2-4x+1 & \Rightarrow u' = (x^2-4x+1)'_x = 2x-4 \\ v' = e^{-x} & \Rightarrow v = \int v'dx = \int e^{-x}dx = -e^{-x} \end{bmatrix} =$$

$$= (x^2-4x+1)\cdot(-e^{-x}) - \int -e^{-x}(2x-4)\,dx =$$

$$= (x^2-4x+1)\cdot(-e^{-x}) + \int \frac{1}{e^x}\cdot(2x-4)\,dx =$$

$$= (x^2-4x+1)\cdot(-e^{-x}) + 2\int \frac{x}{e^x}\,dx - 4\int \frac{dx}{e^x} =$$

$$= \Bigg[\ 1)\ \int \frac{x\,dx}{e^x} = \int x\cdot e^{-x}dx = \begin{bmatrix} u = x & \Rightarrow u' = (x)'_x = 1 \\ v' = e^{-x} & \Rightarrow v = \int v'dx = \int e^{-x}dx = -e^{-x} \end{bmatrix} =$$

$$= x\cdot(-e^{-x}) - \int -e^{-x}dx = x(-e^{-x}) + \int e^{-x}dx =$$

$$= -x\cdot e^{-x} - e^{-x} + C$$

$$2)\ \int e^{-x}dx = -e^{-x} + C\ \Bigg] =$$

$$= (x^2-4x+1)\cdot(-e^{-x}) - 2xe^{-x} - 2e^{-x} + 4e^{-x} + C =$$

$$= -x^2e^{-x} + 4xe^{-x} - e^{-x} - 2xe^{-x} - 2e^{-x} + 4e^{-x} + C =$$

$$= -x^2e^{-x} + 2xe^{-x} + e^{-x} + C$$

8.2.75

$$\int x^3e^x dx = \begin{bmatrix} u = x^3 & \Rightarrow u' = (x^3)' = 3x^2 \\ v' = e^x & \Rightarrow v = \int v'dx = \int e^x dx = e^x \end{bmatrix} =$$

$$= x^3\cdot e^x - \int e^x\cdot 3x^2dx = x^3e^x - 3\int x^2e^x dx = \begin{bmatrix} u = x^2 & \Rightarrow u' = (x^2)' = 2x \\ v' = e^x & \Rightarrow v = \int e^x dx = e^x \end{bmatrix} =$$

$$= x^3e^x - 3\left(x^2\cdot e^x - \int e^x\, 2x\, dx\right) = x^3e^x - 3x^2e^x + 6\int xe^x dx =$$

$$= \begin{bmatrix} u = x & \Rightarrow u' = (x)' = 1 \\ v' = e^x & \Rightarrow v = \int v'dx = \int e^x dx = e^x \end{bmatrix} = x^3e^x - 3x^2e^x + 6\left(xe^x - \int e^x dx\right) =$$

$$= x^3e^x - 3x^2e^x + 6xe^x - 6e^x + C$$

8.2.76

$$\int \frac{\arccos x\,dx}{\sqrt{1+x}} = \left[\begin{array}{l} u = \arccos x \Rightarrow u' = (\arccos x)' = -\frac{1}{\sqrt{1-x^2}} \\ v' = \frac{1}{\sqrt{1+x}} \Rightarrow v = \int v'dx = \int \frac{dx}{\sqrt{1+x}} = \left[\begin{array}{l}\int \frac{dx}{\sqrt{x}} = 2\sqrt{x} \\ \text{сл. 5, где } a=1\end{array}\right] = \\ = 2\sqrt{x+1} \end{array}\right] = \arccos x \cdot 2\sqrt{x+1} - \int -\frac{2\sqrt{1+x}}{\sqrt{1-x^2}}\,dx =$$

$$= 2\sqrt{x+1}\arccos x + 2\int \frac{\sqrt{1+x}}{\sqrt{(1-x)(1+x)}}\,dx = 2\sqrt{x+1}\arccos x + 2\int \frac{dx}{\sqrt{1-x}} =$$

$$= \left[\begin{array}{l}\int \frac{dx}{\sqrt{x}} = 2\sqrt{x} \\ \int f(ax+b)dx = \frac{1}{a}F(ax+b) + C\end{array}\right] = 2\sqrt{x+1}\arccos x - 4\sqrt{1-x} + C$$

8.2.77

$$\int \frac{\arcsin\sqrt{x}}{\sqrt{1-x}}\,dx = \left[\begin{array}{l} u = \arcsin\sqrt{x} \Rightarrow u' = (\arcsin\sqrt{x})' = \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{-1}{2\sqrt{x}} = \frac{1}{2\sqrt{x}\sqrt{1-x}} \\ v' = \frac{1}{\sqrt{1-x}} \Rightarrow v = \int v'dx = \int \frac{dx}{\sqrt{1-x}} = \left[\begin{array}{l}\int \frac{dx}{\sqrt{x}} = 2\sqrt{x} \\ \text{сл. 5, где } a=-1\end{array}\right] = -2\sqrt{1-x}\end{array}\right] =$$

$$= \arcsin\sqrt{x} \cdot \left(-2\sqrt{1-x}\right) - \int -\frac{2\sqrt{1-x}}{2\sqrt{x}\sqrt{1-x}}\,dx = \arcsin\sqrt{x}\left(-2\sqrt{1-x}\right) + \int \frac{dx}{\sqrt{x}} =$$

$$= -2\sqrt{1-x}\arcsin\sqrt{x} + 2\sqrt{x} + C$$

8.2.78

$$\int \frac{x^2dx}{(x^2-1)^2} = \left[\begin{array}{l} u = x \Rightarrow u' = (x)'_x = 1 \\ v' = \frac{x}{(x^2-1)^2} \Rightarrow v = \int v'dx = \int \frac{x}{(x^2-1)^2}\,dx = \left[\begin{array}{l} t = x^2-1 \Rightarrow dt = d(x^2-1) = \\ = (x^2-1)'dx = 2x\,dx\end{array}\right] = \\ = \frac{1}{2}\int \frac{dt}{t^2} = \frac{1}{2}\cdot\left(-\frac{1}{t}\right) = -\frac{1}{2\cdot(x^2-1)}\end{array}\right] =$$

$$= -\frac{x}{2(x^2-1)} - \int -\frac{dx}{2(x^2-1)} = -\frac{x}{2(x^2-1)} + \frac{1}{2}\int \frac{dx}{x^2-1} =$$

$$= -\frac{x}{2(x^2-1)} + \frac{1}{2}\cdot\frac{1}{2}\cdot\ln\left|\frac{x-1}{x+1}\right| + C =$$

$$= -\frac{x}{2(x^2-1)} + \frac{1}{4}\ln\left|\frac{x-1}{x+1}\right| + C$$

8.2.79

$$\int \cos\ln x\, dx = \begin{bmatrix} t = \ln x \Rightarrow dt = d(\ln x) = (\ln x)'_x dx = \frac{dx}{x} \\ \Rightarrow x = e^{\ln x} = e^t \end{bmatrix} =$$

$$= \int \cos t \cdot e^t\, dt = \begin{bmatrix} u = e^t & \Rightarrow u' = (e^t)'_t = e^t \\ v' = \cos t & \Rightarrow v = \int v' dt = \int \cos t\, dt = \sin t \end{bmatrix} =$$

$$= e^t \cdot \sin t - \int \sin t \cdot e^t\, dt = \begin{bmatrix} u = e^t & \Rightarrow u' = (e^t)'_t = e^t \\ v' = \sin t & \Rightarrow v = \int \sin t\, dt = -\cos t \end{bmatrix} =$$

$$= e^t \sin t - \left(e^t \cdot (-\cos t) - \int -\cos t \cdot e^t\, dt\right) = e^t \sin t + e^t \cdot \cos t - \int \cos t \cdot e^t\, dt =$$

= [тогда

$$\int \cos t \cdot e^t\, dt = e^t \cdot \sin t + e^t \cos t - \int \cos t \cdot e^t\, dt$$

$$\int \cos t \cdot e^t\, dt + \int \cos t \cdot e^t\, dt = \sin t \cdot e^t + \cos t \cdot e^t$$

$$2\int \cos t \cdot e^t\, dt = \sin t \cdot e^t + \cos t \cdot e^t$$

$$\int \cos t \cdot e^t\, dt = \frac{\sin t \cdot e^t}{2} + \frac{\cos t \cdot e^t}{2} \quad \Big] =$$

$$= \frac{\sin t \cdot e^t}{2} + \frac{\cos t \cdot e^t}{2} + C = \frac{\sin \ln x \cdot e^{\ln x}}{2} + \frac{\cos \ln x \cdot e^{\ln x}}{2} + C =$$

$$= \frac{\sin \ln x \cdot x + \cos \ln x \cdot x}{2} + C$$

8.2.80

$$\int e^{3x} \cdot \cos^2 x\, dx = \begin{bmatrix} u = \cos^2 x & \Rightarrow u' = (\cos^2 x)'_x = 2\cos x \cdot (-\sin x) \\ v' = e^{3x} & \Rightarrow v = \int e^{3x} dx = \frac{e^{3x}}{3} \end{bmatrix} =$$

$$= \cos^2 x \cdot \frac{e^{3x}}{3} - \int \frac{e^{3x}}{3} \cdot 2\cos x\, (-\sin x)\, dx = \cos^2 x \cdot \frac{e^{3x}}{3} + \frac{1}{3}\int e^{3x} \cdot \sin 2x\, dx =$$

$$= \begin{bmatrix} u = \sin 2x & \Rightarrow u' = (\sin 2x)'_x = 2\cos 2x \\ v' = e^{3x} & \Rightarrow v = \int e^{3x} dx = \frac{e^{3x}}{3} \end{bmatrix} =$$

$$= \cos^2 x \cdot \frac{e^{3x}}{3} + \frac{1}{3}\left(\sin 2x \cdot \frac{e^{3x}}{3} - \int \frac{e^{3x}}{3} \cdot 2\cos 2x\, dx\right) =$$

$\cancel{= (\cos^2 x)\cdot\left(\frac{e^{3x}}{3}\right)dx}$

$$= \Big[\int e^{3x}\cdot \sin 2x\,dx = \sin 2x \cdot \frac{e^{3x}}{3} - \int \frac{e^{3x}}{3}\cdot 2\cos 2x\,dx =$$

$$= \sin 2x \cdot \frac{e^{3x}}{3} - \frac{2}{3}\int e^{3x}\cdot \cos 2x\,dx = \begin{bmatrix} u=\cos 2x \Rightarrow u' = -2\sin 2x \\ v' = e^{3x} \Rightarrow v = \int e^{3x}dx = \frac{e^{3x}}{3} \end{bmatrix} =$$

$$= \sin 2x \cdot \frac{e^{3x}}{3} - \frac{2}{3}\left(\cos 2x \cdot \frac{e^{3x}}{3} - \int \frac{e^{3x}}{3}(-2\sin 2x)\,dx\right) =$$

$$= \frac{\sin 2x \cdot e^{3x}}{3} - \frac{2}{3}\left(\frac{\cos 2x\cdot e^{3x}}{3} + \frac{2}{3}\int e^{3x}\cdot \sin 2x\,dx\right), \text{ тогда}$$

$$\int e^{3x}\cdot \sin 2x\,dx = \frac{\sin 2x\cdot e^{3x}}{3} - \frac{2}{3}\left(\frac{\cos 2x\cdot e^{3x}}{3} + \frac{2}{3}\int e^{3x}\cdot \sin 2x\,dx\right)$$

$$\int e^{3x}\cdot \sin 2x\,dx = \frac{\sin 2x\cdot e^{3x}}{3} - \frac{2\cos 2x\cdot e^{3x}}{9} - \frac{4}{9}\int e^{3x}\cdot \sin 2x\,dx$$

$$\int e^{3x}\cdot \sin 2x\,dx + \frac{4}{9}\int e^{3x}\sin 2x\,dx = \frac{\sin 2x\cdot e^{3x}}{3} - \frac{2\cos 2x\cdot e^{3x}}{9}$$

$$\frac{13}{9}\int e^{3x}\cdot \sin 2x\,dx = \frac{\sin 2x\cdot e^{3x}}{3} - \frac{2\cos 2x\cdot e^{3x}}{9} \qquad =$$

$$\int e^{3x}\cdot \sin 2x\,dx = \frac{3\sin 2x\cdot e^{3x}}{13} - \frac{2\cos 2x\cdot e^{3x}}{13} + C \Big]^{1} =$$

$$= \frac{\cos^2 x\cdot e^{3x}}{3} + \frac{1}{3}\cdot\left(\frac{3\sin 2x\cdot e^{3x}}{13} - \frac{2\cos 2x\cdot e^{3x}}{13}\right) + C =$$

$$= \frac{\cos^2 x\cdot e^{3x}}{3} + \frac{3\sin 2x\cdot e^{3x} - 2\cos 2x\cdot e^{3x}}{39} + C$$

8.2.81 $\int e^{\sqrt{x}}\,dx = \begin{bmatrix} t=\sqrt{x} \Rightarrow dt = d(\sqrt{x}) = (\sqrt{x})'_x\,dx = \frac{dx}{2\sqrt{x}} \\ \Rightarrow \frac{dx}{2\sqrt{x}} = \frac{dx}{2t} \Rightarrow dx = 2t\,dt \end{bmatrix} = \int e^t\, 2t\,dt =$

$$= 2\int t e^t\,dt = \begin{bmatrix} u=t \Rightarrow u' = (t)'_t = 1 \\ v' = e^t \Rightarrow v = \int v'dt = \int e^t dt = e^t \end{bmatrix} = 2\left(te^t - \int e^t dt\right) =$$

$= 2te^t - 2\int e^t dt = 2te^t - 2e^t + C = 2\sqrt{x}\,e^{\sqrt{x}} - 2e^{\sqrt{x}} + C$

8.2.82

$$\int \frac{x\,dx}{\cos^2 x} = \int x \cdot \frac{1}{\cos^2 x}\,dx = \left[\begin{array}{l} u = x \Rightarrow u' = (x)'_x = 1 \\ v' = \frac{1}{\cos^2 x} \Rightarrow v = \int \frac{dx}{\cos^2 x} = \operatorname{tg} x \end{array}\right] =$$

$$= x \cdot \operatorname{tg} x - \int \operatorname{tg} x\,dx = \left[\int \operatorname{tg} x\,dx = -\ln|\cos x| + C\right] =$$

$$= x\operatorname{tg} x + \ln|\cos x| + C$$

8.2.83

$$\int x^3 \cdot e^{x^2} dx = \int x^2 \cdot x e^{x^2} dx = \left[\begin{array}{l} u = x^2 \Rightarrow u' = (x^2)'_x = 2x \\ v' = xe^{x^2} \Rightarrow \int xe^{x^2} dx = \left[t = e^{x^2} \Rightarrow \right.\end{array}\right.$$

$$\Rightarrow dt = d(e^{x^2}) = (e^{x^2})'_x\,dx = e^{x^2} \cdot 2x\,dx \Rightarrow dx = \frac{dt}{2e^{x^2} x}\Big] = \int \frac{xt\,dt}{2tx} = \int \frac{1}{2}\,dt = \frac{t}{2} =$$

$$= \frac{1}{2}e^{x^2}\Big]\Big] = x^2 \cdot \frac{e^{x^2}}{2} - \int 2x \cdot \frac{e^{x^2}}{2}\,dx = x^2 \cdot \frac{e^{x^2}}{2} - \int x \cdot e^{x^2} dx =$$

$$= \left[\int x \cdot e^{x^2} dx = \frac{e^{x^2}}{2} - \text{см. выше}\right] = \frac{x^2 \cdot e^{x^2}}{2} - \frac{e^{x^2}}{2} + C =$$

$$= \frac{x^2 \cdot e^{x^2} - e^{x^2}}{2} + C$$

8.2.84

$$\int \ln(x + \sqrt{x^2+1})\,dx = \left[\begin{array}{l} u = \ln(x + \sqrt{x^2+1}) \Rightarrow u' = \frac{1}{x+\sqrt{x^2+1}}\left(1 + \frac{2x}{2\sqrt{x^2+1}}\right) = \\ v' = 1 \Rightarrow v = \int dx = x \\ = \frac{1}{x+\sqrt{x^2+1}} \cdot \left(1 + \frac{x}{\sqrt{x^2+1}}\right) \end{array}\right] =$$

$$= \ln(x + \sqrt{x^2+1}) \cdot x - \int x \cdot \frac{1}{x + \sqrt{x^2+1}} \cdot \left(1 + \frac{x}{\sqrt{x^2+1}}\right) dx =$$

$$= x\ln(x + \sqrt{x^2+1}) - \int \frac{x}{x+\sqrt{x^2+1}} \cdot \frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1}}\,dx =$$

$$= x\ln(x+\sqrt{x^2+1}) - \int \frac{x\cdot(x+\sqrt{x^2+1})}{(x+\sqrt{x^2+1})\cdot\sqrt{x^2+1}}\,dx =$$

$$= x\ln(x+\sqrt{x^2+1}) - \int \frac{x}{\sqrt{x^2+1}}\,dx = \left[\begin{array}{l} t=\sqrt{x^2+1} \Rightarrow dt = d(\sqrt{x^2+1}) = \\ = (\sqrt{x^2+1})'_x\,dx = \frac{1}{2\sqrt{x^2+1}}\,2x\,dx \Rightarrow \\ \Rightarrow dx = \frac{\sqrt{x^2+1}}{x}\,dt \end{array}\right] =$$

$$= x\ln(x+\sqrt{x^2+1}) - \int dt = x\cdot\ln(x+\sqrt{x^2+1}) - t + C =$$

$$= x\cdot\ln(x+\sqrt{x^2+1}) - \sqrt{x^2+1} + C$$

8.2.85 $\int \sin 2x \cdot \ln\sin x\,dx =$

$$= \left[\begin{array}{l} u=\ln\sin x \Rightarrow u' = (\ln\sin x)'_x = \frac{1}{\sin x}\cdot\cos x \\ v' = \sin 2x \Rightarrow v = \int \sin 2x\,dx = \left[\begin{array}{l} t=2x \Rightarrow dt = d(2x) = \\ = (2x)'_x\,dx = 2dx \Rightarrow dx = \frac{1}{2}dt \end{array}\right] = \frac{1}{2}\int \sin t\,dt = -\frac{\cos t}{2} = \\ \qquad = -\frac{\cos 2x}{2} \end{array}\right]$$

$$= \ln\sin x\cdot\left(-\frac{\cos 2x}{2}\right) - \int -\frac{\cos 2x}{2}\cdot\frac{\cos x}{\sin x}\,dx =$$

$$= -\frac{\cos 2x\cdot\ln\sin x}{2} + \frac{1}{2}\int \frac{\cos 2x\cdot\cos x}{\sin x}\,dx =$$

$$= -\frac{\cos 2x\cdot\ln\sin x}{2} + \frac{1}{2}\int \frac{\cos x\cdot(1-2\sin^2 x)}{\sin x}\,dx =$$

$$= \left[\begin{array}{l} t = 1-2\sin^2 x \Rightarrow dt = d(1-2\sin^2 x) = (1-2\sin^2 x)'_x\,dx = \\ = -4\cos x\sin x\,dx \Rightarrow dx = -\frac{1}{4\cos x\cdot\sin x}\,dt \end{array}\right] \Rightarrow$$

$$\Rightarrow [t = 1-2\sin^2 x \Rightarrow -4\sin^2(x) = 2t-2] =$$

$$= -\frac{\cos 2x\cdot\ln\sin x}{2} + \frac{1}{2}\int \frac{t}{2t-2}\,dt = \left[\begin{array}{l} y = 2t-2 \Rightarrow dy = d(2t-2) = (2t-2)'_t\,dt = \\ = 2dt \Rightarrow dt = \frac{1}{2}dy \end{array}\right]$$

$$\Rightarrow \left[y = 2t-2 \Rightarrow t = \frac{y+2}{2}\right] =$$

$$= -\frac{\cos 2x \cdot \ln \sin x}{2} + \frac{1}{2}\int \frac{y+2}{2y}\cdot\frac{1}{2}\,dy = -\frac{\cos 2x \cdot \ln\sin x}{2} + \frac{1}{8}\int\frac{y+2}{y}\,dy =$$

$$= -\frac{\cos 2x \cdot \ln \sin x}{2} + \frac{1}{8}\int 1+\frac{2}{y}\,dy = -\frac{\cos 2x \cdot \ln \sin x}{2} + \frac{1}{8}\left(\int dy + 2\int\frac{dy}{y}\right) =$$

$$= -\frac{\cos 2x \cdot \ln \sin x}{2} + \frac{1}{8}\left(y + 2\ln|y|\right) + C =$$

$$= -\frac{\cos 2x \cdot \ln \sin x}{2} + \frac{y}{8} + \frac{\ln|y|}{4} + C =$$

$$= -\frac{\cos 2x \cdot \ln \sin x}{2} + \frac{2t-2}{8} + \frac{\ln|2t-2|}{4} + C =$$

$$= -\frac{\cos 2x \cdot \ln \sin x}{2} + \frac{t-1}{4} + \frac{\ln|2t-2|}{4} + C =$$

$$= -\frac{\cos 2x \cdot \ln \sin x}{2} + \frac{1}{4} - \frac{\sin^2 x}{2} - \frac{1}{4} + \frac{\ln|-4\sin^2 x|}{4} + C =$$

$$= -\frac{\cos 2x \cdot \ln \sin x}{2} - \frac{\sin^2 x}{2} + \frac{\ln(4\sin^2 x)}{4} + C =$$

$$= -\frac{\cos 2x \cdot \ln \sin x}{2} - \frac{\sin^2 x}{2} + \frac{\ln 4}{4} + \frac{2\ln \sin x}{4} + C =$$

$$= -\frac{\cos 2x \cdot \ln \sin x}{2} - \frac{\sin^2 x}{2} + \frac{\ln \sin x}{2} + C$$

8.2.86.

$$\int x^2 \arccos 3x\,dx = \left[\begin{matrix} u = \arccos 3x \Rightarrow u' = -\frac{1}{\sqrt{1-(3x)^2}}\cdot 3 \\ v' = x^2 \Rightarrow v = \int x^2\,dx = x^3/3 \end{matrix}\right] =$$

$$= \arccos 3x \cdot \frac{x^3}{3} - \int \frac{x^3}{3}\left(-\frac{3}{\sqrt{1-(3x)^2}}\right)dx =$$

$$= \frac{\arccos 3x \cdot x^3}{3} + \int \frac{x^3}{\sqrt{1-9x^2}}\,dx =$$

$$= \left[\begin{array}{l} t = 1-9x^2 \Rightarrow dt = d(1-9x^2) = (1-9x^2)'_x\,dx = -18x\,dx \Rightarrow dx = -\frac{dt}{18x} \\ 9x^2 = 1-t \Rightarrow x^2 = \frac{1-t}{9} \end{array}\right] =$$

$$= \frac{\arccos 3x \cdot x^3}{3} + \int -\frac{1-t}{162\sqrt{t}}\,dt = \frac{\arccos 3x \cdot x^3}{3} + \int \frac{t-1}{162\sqrt{t}}\,dt =$$

$$= \frac{\arccos 3x \cdot x^3}{3} + \frac{1}{162} \cdot \int \left(\frac{t}{\sqrt{t}} - \frac{1}{\sqrt{t}}\right) dt =$$

$$= \frac{\arccos 3x \cdot x^3}{3} + \frac{1}{162} \cdot \left(\int t^{\frac{1}{2}}\,dt - \int \frac{dt}{\sqrt{t}}\right) =$$

$$= \frac{\arccos 3x \cdot x^3}{3} + \frac{1}{162} \cdot \left(\frac{2t\sqrt{t}}{3} - 2\sqrt{t}\right) + C =$$

$$= \frac{\arccos 3x \cdot x^3}{3} + \frac{t\sqrt{t}}{243} - \frac{\sqrt{t}}{81} + C =$$

$$= \frac{\arccos 3x \cdot x^3}{3} + \frac{(1-9x^2)\cdot\sqrt{1-9x^2}}{243} - \frac{\sqrt{1-9x^2}}{81} + C$$

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$$\int x \cdot \sin\sqrt{x}\,dx = \left[\begin{array}{l} t = \sqrt{x} \Rightarrow dt = d(\sqrt{x}) = (\sqrt{x})'_x\,dx = \frac{1}{2\sqrt{x}}\,dx \Rightarrow dx = 2\sqrt{x}\,dt \\ = 2t\,dt \\ t = \sqrt{x} \Rightarrow x = t^2 \end{array}\right]$$

$$= \int t^2 \cdot \sin t \cdot 2t\,dt = 2\int t^3 \sin t\,dt = \left[\begin{array}{l} u = t^3 \Rightarrow u' = (t^3)'_t = 3t^2 \\ v' = \sin t \Rightarrow v = \int \sin t\,dt = -\cos t \end{array}\right] =$$

$$= 2\left(t^3 \cdot (-\cos t) - \int(-\cos t) \cdot 3t^2\,dt\right) = 2(-t^3\cos t + 3\int \cos t \cdot t^2\,dt =$$

$$= -2t^3 \cdot \cos t + 6\int \cos t \cdot t^2\,dt = \left[\begin{array}{l} u = t^2 \Rightarrow u' = (t^2)'_t = 2t \\ v' = \cos t \Rightarrow v = \int \cos t\,dt = \sin t \end{array}\right] =$$

$= -2t^3 \cdot \cos t + 6\left(t^2 \cdot \sin t - \int \sin t \cdot 2t\, dt\right) = -2t^3 \cos t + 6t^2 \sin t - 12\int \sin t \cdot t\, dt =$

$= \begin{bmatrix} u = t & \Rightarrow u' = (t)'_t = 1 \\ v' = \sin t & \Rightarrow v = \int \sin t\, dt = -\cos t \end{bmatrix} = -2t^3 \cos t + 6t^2 \sin t - 12\left(t\,(-\cos t) - \int -\cos t\, dt\right) =$

$= -2t^3 \cos t + 6t^2 \cdot \sin t + 12t \cdot \cos t - 12\int \cos t\, dt =$

$= -2t^3 \cos t + 6t^2 \cdot \sin t + 12t \cdot \cos t - 12\sin t + C =$

$= -2(\sqrt{x})^3 \cos\sqrt{x} + 6(\sqrt{x})^2 \cdot \sin\sqrt{x} + 12\sqrt{x} \cdot \cos\sqrt{x} - 12\sin\sqrt{x} + C =$

$= -2x\sqrt{x} \cdot \cos\sqrt{x} + 6x \cdot \sin\sqrt{x} + 12\sqrt{x} \cdot \cos\sqrt{x} - 12\sin\sqrt{x} + C.$

8.2.88

$\int \arcsin^2 x\, dx = \begin{bmatrix} u = \arcsin^2 x & \Rightarrow u' = (\arcsin^2 x)'_x = 2\arcsin x \cdot \frac{1}{\sqrt{1-x^2}} \\ v' = 1 & \Rightarrow v = \int dx = x \end{bmatrix} =$

$= \arcsin^2 x \cdot x - \int x \cdot 2\arcsin x \cdot \frac{1}{\sqrt{1-x^2}} \cdot dx =$

$= \arcsin^2 x \cdot x - 2\int \arcsin x \cdot \frac{x}{\sqrt{1-x^2}}\, dx \overset{1}{=} \left[\begin{matrix} u = \arcsin x & \Rightarrow u' = \frac{1}{\sqrt{1-x^2}} \\ v' = \frac{x}{\sqrt{1-x^2}} & \Rightarrow v = \int \frac{x}{\sqrt{1-x^2}}\, dx = \end{matrix}\right.$

$\overset{2}{=} \left[\ \int \frac{x}{\sqrt{1-x^2}}\, dx \overset{3}{=} \left[\begin{matrix} t = \sqrt{1-x^2} \Rightarrow dt = d(\sqrt{1-x^2}) = (\sqrt{1-x^2})'_x\, dx = \frac{1}{2\sqrt{1-x^2}} \cdot (2x)\, dx \overset{3}{\Rightarrow} \\ \Rightarrow -\frac{\sqrt{1-x^2}}{x}\, dt = dx \end{matrix}\right] = \right.$

$= \int \frac{x}{t} \cdot \left(-\frac{t}{x}\right) dt = \int -dt = -t + C = -\sqrt{1-x^2} + C \Big]^2 = -\sqrt{1-x^2} \Big]$

$= \arcsin^2 x \cdot x - 2\left(\arcsin x \cdot (-\sqrt{1-x^2}) - \int(-\sqrt{1-x^2}) \cdot \frac{1}{\sqrt{1-x^2}}\, dx\right) =$

$= \arcsin^2 x \cdot x + 2\sqrt{1-x^2}\arcsin x - 2\int dx =$

$= x\arcsin^2 x + 2\sqrt{1-x^2}\arcsin x - 2x + C$

8.2.89. $\int \frac{\cos\sqrt{x}}{\sqrt{x}}dx = \left[\begin{array}{l} t=\sqrt{x} \Rightarrow dt = d(\sqrt{x}) = (\sqrt{x})'_x dx = \frac{1}{2\sqrt{x}}dx \Rightarrow \\ \Rightarrow dx = 2t\,dt \end{array}\right] =$

$= \int \frac{\cos t \cdot 2t}{t}dt = \int 2\cos t\,dt = 2\int \cos t\,dt = 2\sin t + C =$

$= 2\sin\sqrt{x} + C$

8.2.90 $\int \operatorname{arctg}\sqrt{x}\,dx = \left[\begin{array}{l} t=\sqrt{x} \Rightarrow dt = \frac{1}{2\sqrt{x}}dx \Rightarrow \\ \Rightarrow dx = 2t\,dt \end{array}\right] = \int \operatorname{arctg} t \cdot 2t\,dt =$

$= 2\int \operatorname{arctg} t \cdot t\,dt = \left[\begin{array}{l} u = \operatorname{arctg} t \Rightarrow u' = (\operatorname{arctg} t)'_t = \frac{1}{1+t^2} \\ v' = t \Rightarrow v = \int t\,dt = \frac{t^2}{2} \end{array}\right] =$

$= 2\left(\operatorname{arctg} t \cdot \frac{t^2}{2} - \int \frac{t^2}{2} \cdot \frac{1}{1+t^2}dt\right) = 2\left(\operatorname{arctg} t \cdot \frac{t^2}{2} - \frac{1}{2}\int \frac{t^2}{1+t^2}dt\right) =$

$= \operatorname{arctg} t \cdot t^2 - \int \frac{t^2+1-1}{1+t^2}dt = t^2 \cdot \operatorname{arctg} t - \int\left(\frac{1+t^2}{1+t^2} - \frac{1}{1+t^2}\right)dt =$

$= t^2 \cdot \operatorname{arctg} t - \left(\int dt - \int \frac{dt}{1+t^2}\right) = t^2 \operatorname{arctg} t - t + \operatorname{arctg} t + C =$

$= x \cdot \operatorname{arctg}\sqrt{x} - \sqrt{x} + \operatorname{arctg}\sqrt{x} + C.$

8.2.91. $\int \frac{dx}{\sin x} = \int \frac{1}{\sin x} \cdot \frac{\sin x}{\sin x}dx = \int \frac{\sin x}{\sin^2 x}dx = \int \frac{\sin x}{1-\cos^2 x}dx =$

$= \left[\begin{array}{l} t = \cos x \Rightarrow dt = d(\cos x) = (\cos x)'_x dx = -\sin x\,dx \Rightarrow \\ \Rightarrow dx = -\frac{1}{\sin x}dt \end{array}\right] = \int -\frac{\sin x}{1-t^2} \cdot \frac{1}{\sin x}dt =$

$= \int -\frac{dt}{1-t^2} = \int \frac{dt}{-1+t^2} = \int \frac{dt}{t^2-1^2} = \left[\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C,\ (a>0)\right] =$

$= \frac{1}{2}\ln\left|\frac{t-1}{t+1}\right| + C = \frac{1}{2}\ln\left|\frac{\cos x-1}{\cos x+1}\right| + C = \frac{1}{2}\ln\left|-\operatorname{tg}^2\frac{x}{2}\right| + C =$

$$= \ln\left(tg^2\frac{x}{2}\right)^{\frac{1}{2}} + C = \ln\left(tg\frac{x}{2}\right) + C$$

8.2.92

$$\int \frac{\ln^2 x}{x\cdot\sqrt{3-\ln x}}\,dx = \left[\begin{array}{l} t=\ln x \Rightarrow dt = d(\ln x) = \\ = (\ln x)'_x\,dx = \frac{1}{x}dx \end{array}\right] = \int \frac{t^2}{\sqrt{3-t}}\,dt =$$

$$= \left[\begin{array}{l} y = 3-t \Rightarrow dy = d(3-t) = (3-t)'_t\,dt = -dt \\ y = 3-t \Rightarrow t = 3-y \end{array}\right] = \int -\frac{(3-y)^2}{\sqrt{y}}\,dy = -\int \frac{9-6y+y^2}{\sqrt{y}}\,dy =$$

$$= -\int\left(\frac{9}{\sqrt{y}} - \frac{6y}{\sqrt{y}} + \frac{y^2}{\sqrt{y}}\right)dy = -\left(\int\frac{9}{\sqrt{y}}\,dy - \int\frac{6y}{\sqrt{y}}\,dy + \int\frac{y^2}{\sqrt{y}}\,dy\right) =$$

$$= -9\int\frac{dy}{\sqrt{y}} + 6\int y^{\frac{1}{2}}\,dy - \int y^{\frac{3}{2}}\,dy = \left[\begin{array}{l} 1)\int\frac{dx}{\sqrt{x}} = 2\sqrt{x} + C \\ 2)\ x^a dx = \frac{x^{a+1}}{a+1} + C\ (a \neq -1) \end{array}\right] =$$

$$= -18\sqrt{y} + 4y\sqrt{y} - \frac{2y^2\sqrt{y}}{5} + C =$$

$$= -18\sqrt{3-t} + 4(3-t)\sqrt{3-t} - \frac{2(3-t)^2\sqrt{3-t}}{5} + C =$$

$$= -18\sqrt{3-t} + 4(3-t)\sqrt{3-t} - \frac{2(9-6t+t^2)\sqrt{3-t}}{5} + C =$$

$$= -18\sqrt{3-\ln x} + 4(3-\ln x)\sqrt{3-\ln x} - \frac{2(9-6\ln x+\ln^2 x)\sqrt{3-\ln x}}{5} + C$$

8.2.93

$$\int\frac{e^{arctg x}+8x}{1+x^2}\,dx = \int\left(\frac{e^{arctg x}}{1+x^2} + \frac{8x}{1+x^2}\right)dx = \int\frac{e^{arctg x}}{1+x^2}\,dx + 8\int\frac{x}{1+x^2}\,dx =$$

$$= \left[\begin{array}{l} 1)\ t = e^{arctg x} \Rightarrow dt = d(e^{arctg x}) = (e^{arctg x})'_x\,dx = e^{arctg x}\cdot\frac{1}{x^2+1}\,dx \\ 2)\ y = 1+x^2 \Rightarrow dy = d(1+x^2) = (1+x^2)'_x\,dx = 2x\,dx \end{array}\right] =$$

$$= \int dt + 8\int\frac{1}{2y}\,dy = \int dt + 4\int\frac{dy}{y} = t + 4\ln|y| + C =$$

$$= e^{\operatorname{arctg} x} + 4\ln|1+x^2| + C = [1+x^2>0] =$$

$$= e^{\operatorname{arctg} x} + 4\ln(1+x^2) + C$$

8.2.94

$$\int \frac{3x + 5\sin\frac{1}{e^x}}{e^x}\,dx = 3\int \frac{x}{e^x}\,dx + 5\int \frac{\sin\frac{1}{e^x}}{e^x}\,dx =$$

$$= \Big[\ 1)\ \int \frac{x}{e^x}\,dx = \left[\begin{array}{ll} u = x & \Rightarrow u' = (x)'_x = 1 \\ v' = \frac{1}{e^x} & \Rightarrow v = \int \frac{1}{e^x}\,dx = -\frac{1}{e^x} \end{array}\right] = x\cdot\left(-\frac{1}{e^x}\right) - \int -\frac{1}{e^x}\,dx =$$

$$= -\frac{x}{e^x} + \int \frac{1}{e^x}\,dx = -\frac{x}{e^x} - \frac{1}{e^x} + C = -\frac{x+1}{e^x} + C$$

$$2)\ \int \frac{\sin\frac{1}{e^x}}{e^x}\,dx = \left[\begin{array}{l} t = \frac{1}{e^x} \Rightarrow dt = d\left(\frac{1}{e^x}\right) = \left(\frac{1}{e^x}\right)'_x dx = \\ = -\frac{e^x}{(e^x)^2}\,dx = -\frac{1}{e^x}\,dx \end{array}\right] = \int -\sin t\,dt = -\int \sin t\,dt =$$

$$= \cos t + C = \cos\frac{1}{e^x} + C\ \Big] = 3\cdot\left(-\frac{x+1}{e^x}\right) + 5\cdot\cos\frac{1}{e^x} + C =$$

$$= 5\cos\frac{1}{e^x} - \frac{3x+3}{e^x} + C$$